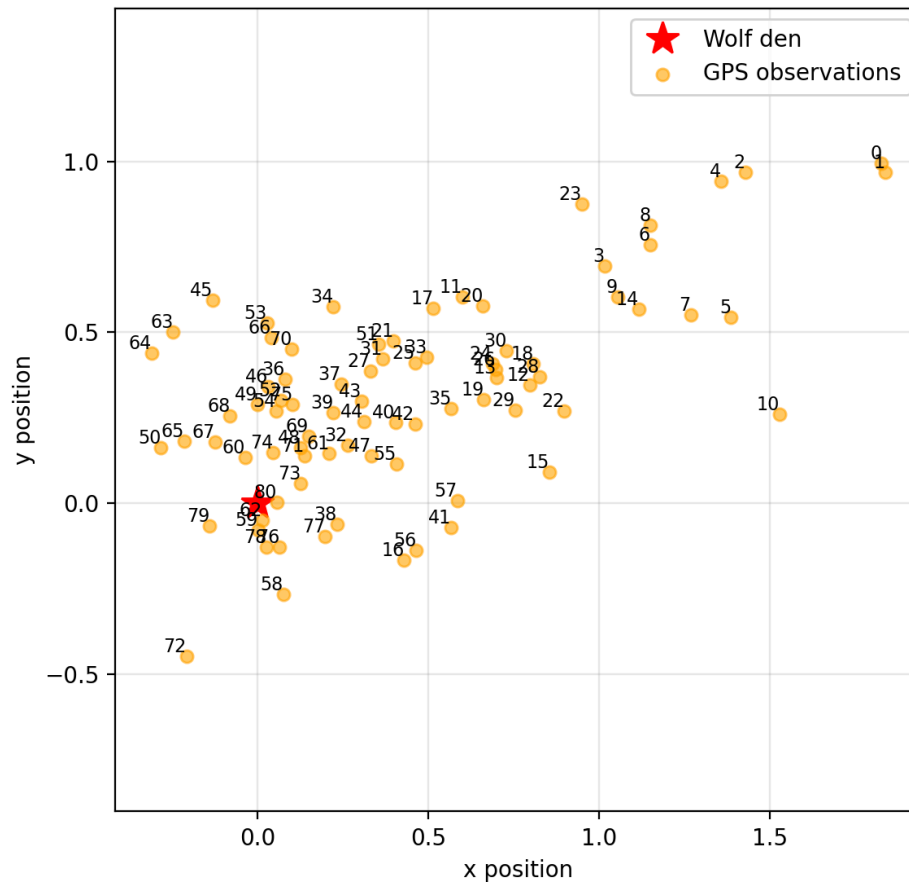


Foundations of Data Science

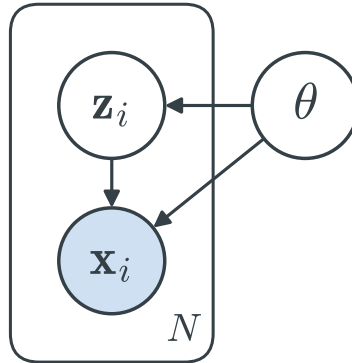
Lecture 5: State-space models

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Today's case study: **tracking** the location of a wolf over time from noisy GPS observations, here simulated.

Discrete-time models



Static latent variable models

We previously defined latent variable models as probabilistic models that explain observed data $\mathbf{x}$ in terms of unobserved (latent) variables $\mathbf{z}$ and parameters θ ,

$$p(\mathbf{x}, \mathbf{z}, \theta) = p(\mathbf{x}|\mathbf{z}, \theta)p(\mathbf{z}|\theta)p(\theta).$$

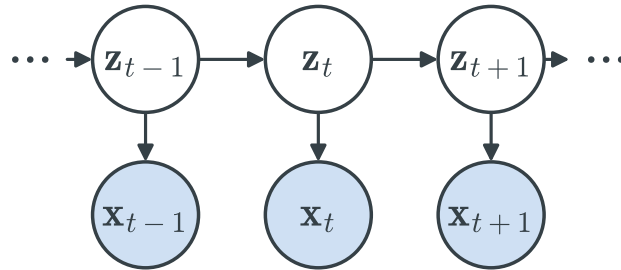


What if the system evolves over time and we have a sequence of observations $\mathbf{x}_{1:T} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ collected at discrete time steps $t = 1, \dots, T$?

State-space models

To model a dynamical system, we can first assume a discretization of time and introduce a sequence of latent variables $\mathbf{z}_t$ that represent the state of the system at each time step t .

In this context, a **state-space model** is a latent variable model that explains a sequence $\mathbf{x}_{1:T} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ of observations in terms of a sequence $\mathbf{z}_{1:T} = (\mathbf{z}_1, \dots, \mathbf{z}_T)$ of latent variables.



In a Markovian state-space model, the latent variables form a Markov chain

$$p(\mathbf{z}_t | \mathbf{z}_{1:t-1}) = p(\mathbf{z}_t | \mathbf{z}_{t-1}),$$

where the conditional distribution $p(\mathbf{z}_t | \mathbf{z}_{t-1})$ is called the **transition model**.

The observations are conditionally independent given the latent variables,

$$p(\mathbf{x}_t | \mathbf{x}_{1:t-1}, \mathbf{z}_{1:t}) = p(\mathbf{x}_t | \mathbf{z}_t),$$

where the conditional distribution $p(\mathbf{x}_t | \mathbf{z}_t)$ is called the **observation model**.



Example

We want to track the location of a wild animal (e.g., a wolf) over time using noisy GPS observations.

Assumptions:

- The animal has a home location (den, nest) at $\mu \in \mathbb{R}^2$.
- The animal moves according to a random walk with drift towards the home location.
- The GPS observations are noisy measurements of the animal's true location.
- Time is discretized regularly every Δt time units.

State-space model:

- States $\mathbf{z}_t \in \mathbb{R}^2$ represent the true location of the animal at time t . They evolve as a random walk with drift towards the home location $\boldsymbol{\mu}$.

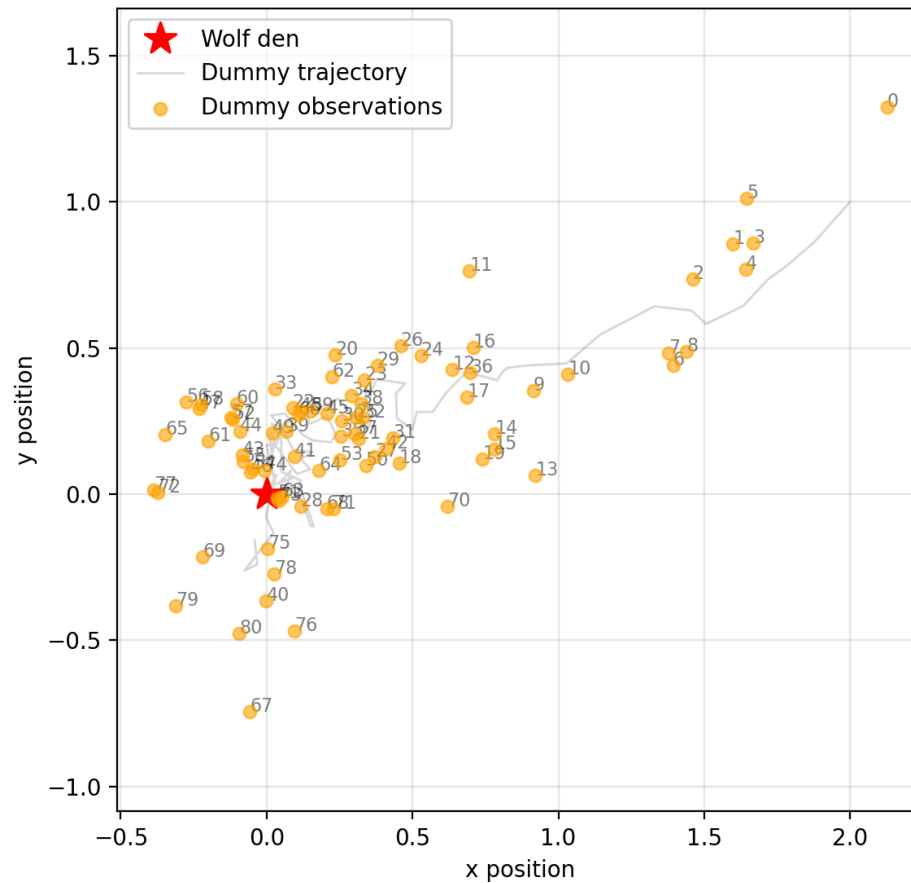
$$p(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t | \mathbf{z}_{t-1} - \kappa(\mathbf{z}_{t-1} - \boldsymbol{\mu})\Delta t, \sigma^2 \Delta t \mathbf{I}),$$

where $\kappa > 0$ is the strength of attraction to the home location and σ^2 is the variance of the random walk.

- Observations $\mathbf{x}_t \in \mathbb{R}^2$ represent noisy GPS measurements of the animal's location at time t . They relate to the states via

$$p(\mathbf{x}_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{x}_t | \mathbf{z}_t, \mathbf{R}),$$

where $\mathbf{R}$ is the observation noise covariance.



Example of trajectory and observations
generated from the discrete-time state-space model ($\Delta t = 0.25$).

Inference in state-space models

Given a state-space model and a sequence of observations $\mathbf{x}_{1:T}$, we are typically interested in solving one or more of the following inference problems:

- Prediction: $p(\mathbf{z}_{t+k} | \mathbf{x}_{1:t})$ for $k \geq 1$.
- Filtering: $p(\mathbf{z}_t | \mathbf{x}_{1:t})$.
- Smoothing: $p(\mathbf{z}_t | \mathbf{x}_{1:T})$.

Bayes filter

The Bayes filter is a recursive algorithm for estimating the filtering distributions $p(\mathbf{z}_t | \mathbf{x}_{1:t})$ as

$$p(\mathbf{z}_t | \mathbf{x}_{1:t}) = \frac{p(\mathbf{x}_t | \mathbf{z}_t) \int p(\mathbf{z}_t | \mathbf{z}_{t-1}) p(\mathbf{z}_{t-1} | \mathbf{x}_{1:t-1}) d\mathbf{z}_{t-1}}{p(\mathbf{x}_t | \mathbf{x}_{1:t-1})},$$

for $t = 1, 2, \dots, T$, with the base case $p(\mathbf{z}_1 | \mathbf{x}_1) = \frac{p(\mathbf{x}_1 | \mathbf{z}_1) p(\mathbf{z}_1)}{p(\mathbf{x}_1)}$.

Proof. The filtering distribution can be derived in two steps:

1. Prediction: Push the filtering distribution from the previous time step through the transition model to obtain a prediction of the current state. That is,

$$p(\mathbf{z}_t | \mathbf{x}_{1:t-1}) = \int p(\mathbf{z}_t | \mathbf{z}_{t-1}) p(\mathbf{z}_{t-1} | \mathbf{x}_{1:t-1}) d\mathbf{z}_{t-1}.$$

2. Update: Condition on the new observation to obtain the filtering distribution,

$$p(\mathbf{z}_t | \mathbf{x}_{1:t}) = \frac{p(\mathbf{x}_t | \mathbf{z}_t) p(\mathbf{z}_t | \mathbf{x}_{1:t-1})}{p(\mathbf{x}_t | \mathbf{x}_{1:t-1})},$$

where the marginal likelihood $p(\mathbf{x}_t | \mathbf{x}_{1:t-1})$ is given by

$$p(\mathbf{x}_t | \mathbf{x}_{1:t-1}) = \int p(\mathbf{x}_t | \mathbf{z}_t) p(\mathbf{z}_t | \mathbf{x}_{1:t-1}) d\mathbf{z}_t.$$

Once we have computed the filtering distributions $p(\mathbf{z}_t | \mathbf{x}_{1:t})$ for $t = 1, \dots, T$, we can compute the **prediction distributions** $p(\mathbf{z}_{t+k} | \mathbf{x}_{1:t})$ for $k \geq 1$ using the prediction step of the Bayes filter iteratively,

$$p(\mathbf{z}_{t+k} | \mathbf{x}_{1:t}) = \int p(\mathbf{z}_{t+k} | \mathbf{z}_{t+k-1}) p(\mathbf{z}_{t+k-1} | \mathbf{x}_{1:t}) d\mathbf{z}_{t+k-1},$$

for $k = 1, 2, \dots$

Bayes smoother

The Bayes smoother computes the smoothing distributions $p(\mathbf{z}_t | \mathbf{x}_{1:T})$ for $t = 1, \dots, T$ using the filtering distributions $p(\mathbf{z}_t | \mathbf{x}_{1:t})$ and the transition model $p(\mathbf{z}_t | \mathbf{z}_{t-1})$. It consists of a backward recursion,

$$p(\mathbf{z}_t | \mathbf{x}_{1:T}) = p(\mathbf{z}_t | \mathbf{x}_{1:t}) \int \frac{p(\mathbf{z}_{t+1} | \mathbf{z}_t) p(\mathbf{z}_{t+1} | \mathbf{x}_{1:T})}{p(\mathbf{z}_{t+1} | \mathbf{x}_{1:t})} d\mathbf{z}_{t+1},$$

for $t = T - 1, T - 2, \dots, 1$, starting from $p(\mathbf{z}_T | \mathbf{x}_{1:T})$, the filtering distribution at $t = T$.

Proof. The joint distribution $p(\mathbf{z}_t, \mathbf{z}_{t+1} | \mathbf{x}_{1:T})$ can be computed as

$$\begin{aligned} p(\mathbf{z}_t, \mathbf{z}_{t+1} | \mathbf{x}_{1:T}) &= p(\mathbf{z}_t | \mathbf{z}_{t+1}, \mathbf{x}_{1:T}) p(\mathbf{z}_{t+1} | \mathbf{x}_{1:T}) \\ &= p(\mathbf{z}_t | \mathbf{z}_{t+1}, \mathbf{x}_{1:t}) p(\mathbf{z}_{t+1} | \mathbf{x}_{1:T}) \\ &= \frac{p(\mathbf{z}_{t+1} | \mathbf{z}_t) p(\mathbf{z}_t | \mathbf{x}_{1:t})}{p(\mathbf{z}_{t+1} | \mathbf{x}_{1:t})} p(\mathbf{z}_{t+1} | \mathbf{x}_{1:T}), \end{aligned}$$

where we used the conditional independence properties of the state-space model.

Marginalizing over $\mathbf{z}_{t+1}$ gives the desired result,

$$\begin{aligned} p(\mathbf{z}_t | \mathbf{x}_{1:T}) &= \int p(\mathbf{z}_t, \mathbf{z}_{t+1} | \mathbf{x}_{1:T}) d\mathbf{z}_{t+1} \\ &= p(\mathbf{z}_t | \mathbf{x}_{1:t}) \int \frac{p(\mathbf{z}_{t+1} | \mathbf{z}_t) p(\mathbf{z}_{t+1} | \mathbf{x}_{1:T})}{p(\mathbf{z}_{t+1} | \mathbf{x}_{1:t})} d\mathbf{z}_{t+1}. \end{aligned}$$



In general, the integrals of the Bayes filter and smoother have no closed form.

Two classes of models are exceptions:

- linear Gaussian models, where the integrals are Gaussian (Kalman filter and smoother);
- hidden Markov models, where they are finite sums (forward-backward algorithm).

Linear Gaussian state-space models

A linear Gaussian state-space model (LGSSM) is a state-space model where both the transition and observation models are linear Gaussian. That is,

$$\begin{aligned}p(\mathbf{z}_t | \mathbf{z}_{t-1}) &= \mathcal{N}(\mathbf{z}_t | \mathbf{A}\mathbf{z}_{t-1} + \mathbf{b}, \mathbf{Q}), \\p(\mathbf{x}_t | \mathbf{z}_t) &= \mathcal{N}(\mathbf{x}_t | \mathbf{H}\mathbf{z}_t, \mathbf{R}),\end{aligned}$$

where $\mathbf{A}$ is the state transition matrix, $\mathbf{b}$ an offset, $\mathbf{Q}$ the process noise covariance, $\mathbf{H}$ the observation matrix, and $\mathbf{R}$ the observation noise covariance.

The wolf model is of this form, with $\mathbf{A} = (1 - \kappa\Delta t)\mathbf{I}$, $\mathbf{b} = \kappa\Delta t \boldsymbol{\mu}$, $\mathbf{Q} = \sigma^2\Delta t \mathbf{I}$ and $\mathbf{H} = \mathbf{I}$.

If the prior distribution $p(\mathbf{z}_1)$ is also Gaussian, then all filtering, prediction, and smoothing distributions are Gaussian.

The **Kalman filter** provides a closed-form expression for the filtering distributions in linear Gaussian state-space models.

At each time step t , $p(\mathbf{z}_t | \mathbf{x}_{1:t}) = \mathcal{N}(\mathbf{z}_t | \mathbf{m}_t, \mathbf{P}_t)$ is Gaussian with mean $\mathbf{m}_t$ and covariance $\mathbf{P}_t$. These parameters can be computed recursively by adapting the Bayes filter equations to the linear Gaussian case.

Proof. By recursion, assume that at time step $t - 1$, the filtering distribution is Gaussian, $p(\mathbf{z}_{t-1}|\mathbf{x}_{1:t-1}) = \mathcal{N}(\mathbf{z}_{t-1}|\mathbf{m}_{t-1}, \mathbf{P}_{t-1})$, with the base case $p(\mathbf{z}_1|\mathbf{x}_1) = \mathcal{N}(\mathbf{z}_1|\mathbf{m}_1, \mathbf{P}_1)$.

For the prediction step, we have

$$\begin{aligned}
 p(\mathbf{z}_t|\mathbf{x}_{1:t-1}) &= \int p(\mathbf{z}_{t-1}|\mathbf{x}_{1:t-1})p(\mathbf{z}_t|\mathbf{z}_{t-1})d\mathbf{z}_{t-1} \\
 &= \int \mathcal{N}(\mathbf{z}_{t-1}|\mathbf{m}_{t-1}, \mathbf{P}_{t-1})\mathcal{N}(\mathbf{z}_t|\mathbf{A}\mathbf{z}_{t-1} + \mathbf{b}, \mathbf{Q})d\mathbf{z}_{t-1} \\
 &= \int \mathcal{N}\left(\begin{pmatrix} \mathbf{z}_{t-1} \\ \mathbf{z}_t \end{pmatrix} \middle| \begin{bmatrix} \mathbf{m}_{t-1} \\ \mathbf{A}\mathbf{m}_{t-1} + \mathbf{b} \end{bmatrix}, \begin{bmatrix} \mathbf{P}_{t-1} & \mathbf{P}_{t-1}\mathbf{A}^T \\ \mathbf{A}\mathbf{P}_{t-1} & \mathbf{A}\mathbf{P}_{t-1}\mathbf{A}^T + \mathbf{Q} \end{bmatrix}\right) d\mathbf{z}_{t-1} \\
 &= \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t^-, \mathbf{P}_t^-),
 \end{aligned}$$

where $\mathbf{m}_t^- = \mathbf{A}\mathbf{m}_{t-1} + \mathbf{b}$ and $\mathbf{P}_t^- = \mathbf{A}\mathbf{P}_{t-1}\mathbf{A}^T + \mathbf{Q}$.

For the update step, we join the prediction distribution with the observation model,

$$\begin{aligned} p\left(\begin{pmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{pmatrix} \mid \mathbf{x}_{1:t-1}\right) &= p(\mathbf{z}_t \mid \mathbf{x}_{1:t-1})p(\mathbf{x}_t \mid \mathbf{z}_t) \\ &= \mathcal{N}\left(\begin{pmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{pmatrix} \mid \begin{bmatrix} \mathbf{m}_t^- \\ \mathbf{H}\mathbf{m}_t^- \end{bmatrix}, \begin{bmatrix} \mathbf{P}_t^- & \mathbf{P}_t^- \mathbf{H}^T \\ \mathbf{H}\mathbf{P}_t^- & \mathbf{H}\mathbf{P}_t^- \mathbf{H}^T + \mathbf{R} \end{bmatrix}\right). \end{aligned}$$

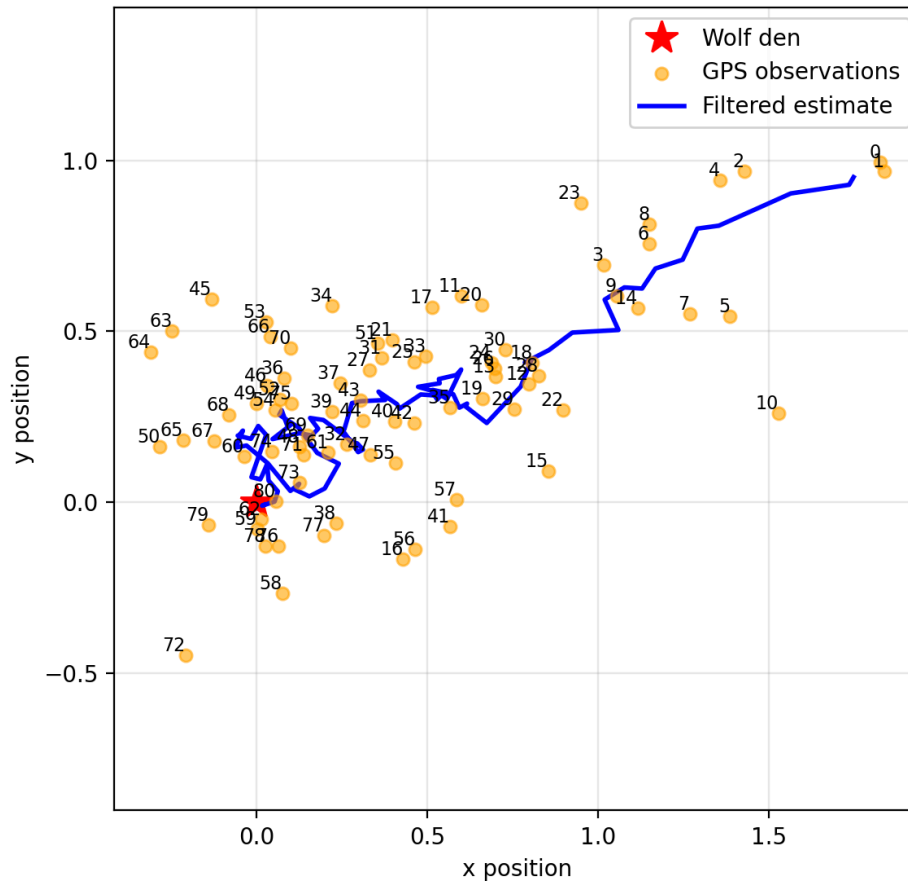
Therefore, the filtering distribution is given by the conditional distribution

$$p(\mathbf{z}_t \mid \mathbf{x}_{1:t-1}, \mathbf{x}_t) = p(\mathbf{z}_t \mid \mathbf{x}_{1:t}) = \mathcal{N}(\mathbf{z}_t \mid \mathbf{m}_t, \mathbf{P}_t),$$

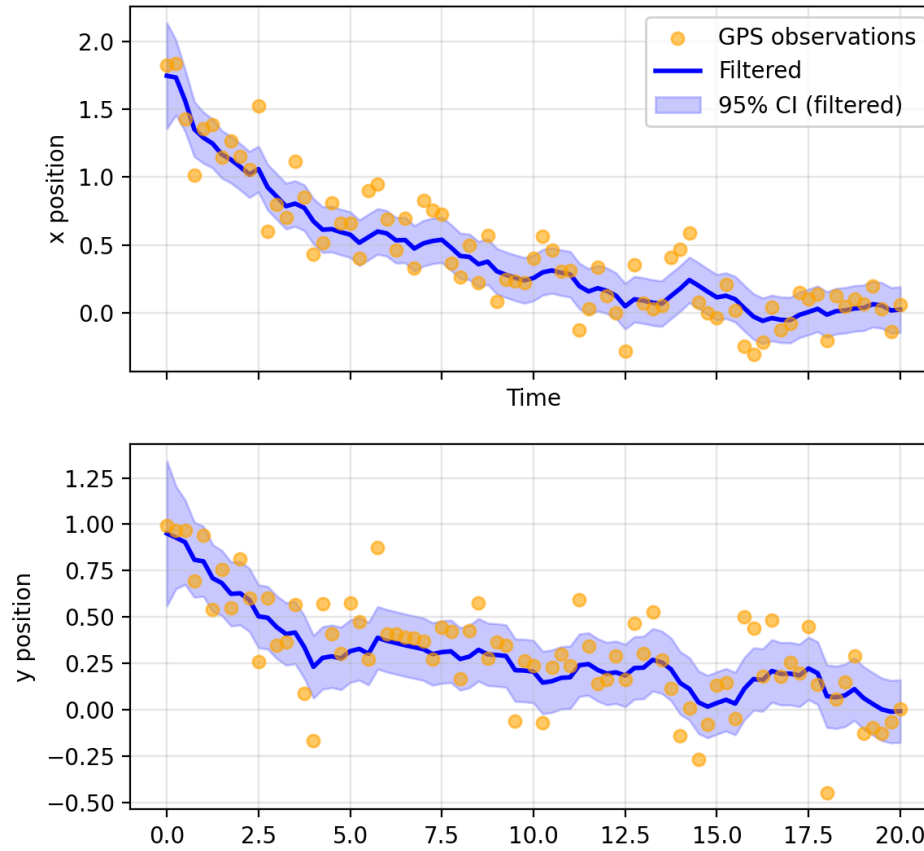
where

$$\begin{aligned} \mathbf{m}_t &= \mathbf{m}_t^- + \mathbf{K}_t(\mathbf{x}_t - \mathbf{H}\mathbf{m}_t^-), \\ \mathbf{P}_t &= (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_t^-, \end{aligned}$$

and $\mathbf{K}_t = \mathbf{P}_t^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_t^- \mathbf{H}^T + \mathbf{R})^{-1}$ is the Kalman gain and represents the weight given to the new observation.



Mean estimate of the wolf's trajectory using the Kalman filter.



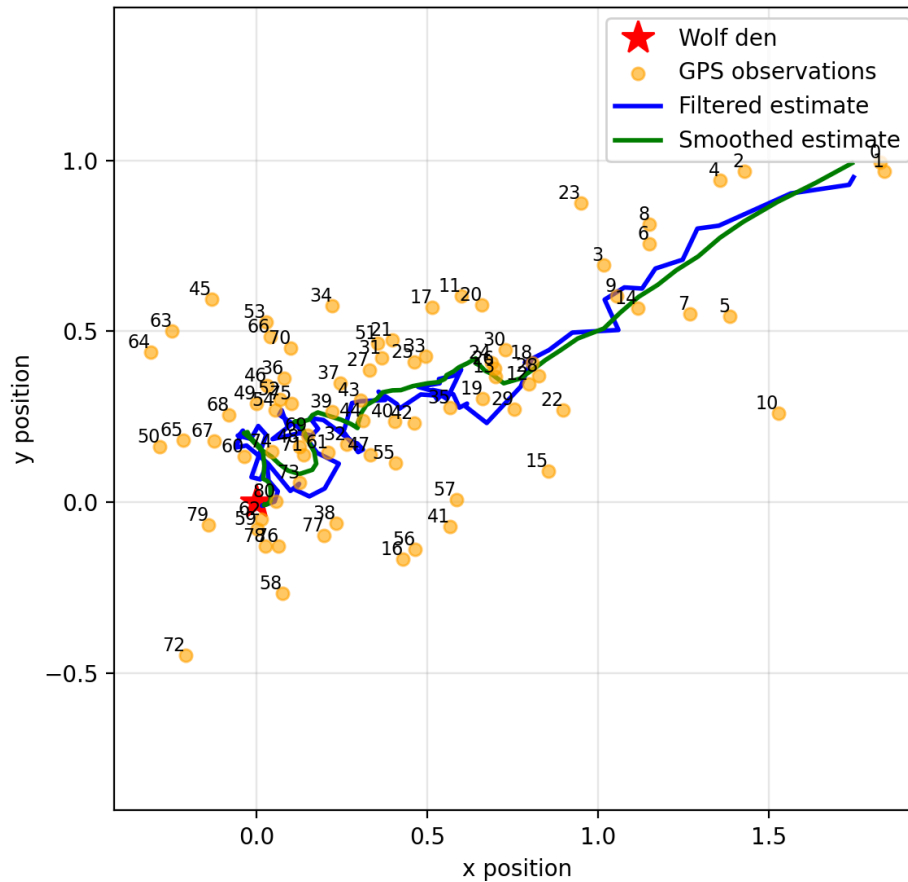
Filtering distribution at each time step using the Kalman filter.

The smoothing distributions $p(\mathbf{z}_t | \mathbf{x}_{1:T}) = \mathcal{N}(\mathbf{z}_t | \mathbf{m}_t^s, \mathbf{P}_t^s)$ are also Gaussian with mean $\mathbf{m}_t^s$ and covariance $\mathbf{P}_t^s$.

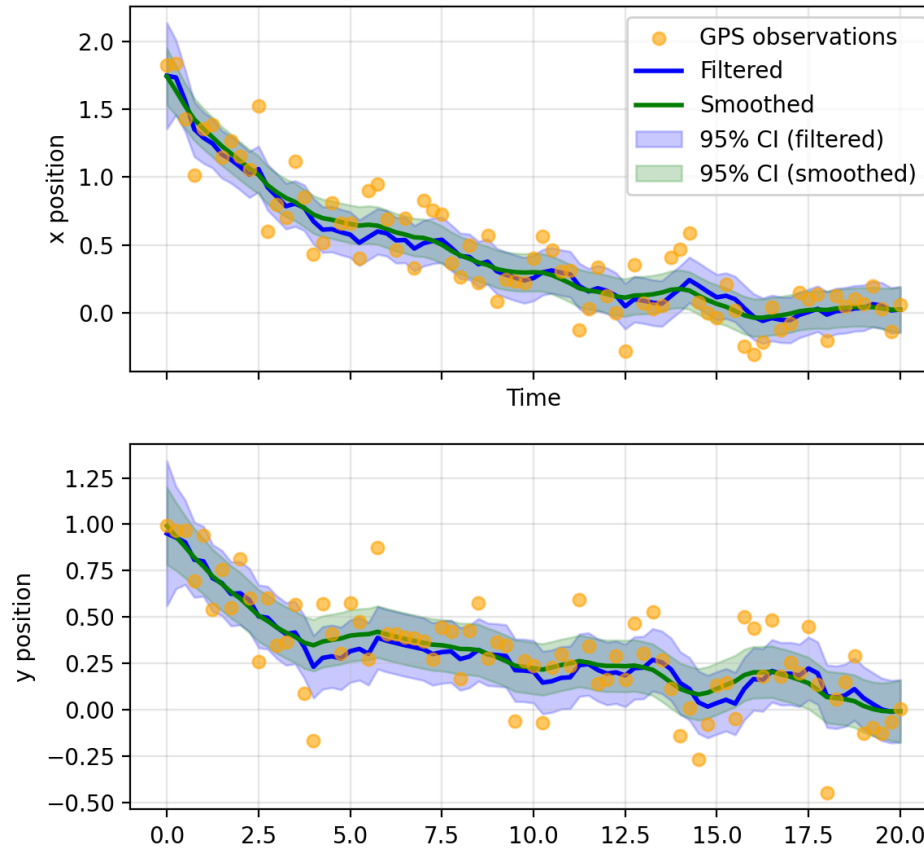
The parameters can be computed recursively using the **Rauch-Tung-Striebel smoother** equations (proof omitted for brevity),

$$\begin{aligned}\mathbf{C}_t &= \mathbf{P}_t \mathbf{A}^T (\mathbf{P}_{t+1}^-)^{-1}, \\ \mathbf{m}_t^s &= \mathbf{m}_t + \mathbf{C}_t (\mathbf{m}_{t+1}^s - \mathbf{m}_{t+1}^-), \\ \mathbf{P}_t^s &= \mathbf{P}_t + \mathbf{C}_t (\mathbf{P}_{t+1}^s - \mathbf{P}_{t+1}^-) \mathbf{C}_t^T,\end{aligned}$$

for $t = T - 1, T - 2, \dots, 1$, with the base case $\mathbf{m}_T^s = \mathbf{m}_T$ and $\mathbf{P}_T^s = \mathbf{P}_T$.



Mean estimate of the wolf's trajectory using the Kalman smoother.



Smoothing distribution at each time step using the Kalman smoother.

Hidden Markov models

A **hidden Markov model** (HMM) is a state-space model whose states are discrete, $z_t \in \{1, \dots, K\}$, with transition probabilities

$$p(z_t = j \mid z_{t-1} = i) = \mathbf{A}_{i,j},$$

and any observation model $p(x_t \mid z_t = j)$ for each state j . It is the mixture model of Lecture 4, with a component that switches over time.

Example. The behavior of a wolf (resting, foraging, traveling) as the state, its speed as the observation. $\mathbf{A}$ says how often the wolf switches from one behavior to another.

With discrete states, the integrals of the Bayes filter become sums over the K states. Writing $\alpha_t(j) = p(z_t = j \mid x_{1:t})$,

$$\alpha_t(j) \propto p(x_t \mid z_t = j) \sum_{i=1}^K \mathbf{A}_{i,j} \alpha_{t-1}(i),$$

normalized over j . This is the **forward algorithm**, at a cost of K^2 operations per step.

A similar backward recursion gives the smoothing distributions: together, the **forward-backward algorithm**.

Continuous-time models



So far, time was discretized with a fixed step Δt , and the transition and observation models were defined at these steps.

Two reasons to model in **continuous time**:

- physical processes are naturally described by rates of change;
- observations may arrive at **irregular times**, triggered by events, or at several time scales.

From discrete to continuous time

In discrete-time state-space models, we considered transition models of the form $p(\mathbf{z}_t | \mathbf{z}_{t-1})$ that describe how the state evolves from one time step to the next. For additive transitions and additive noise, this can be expressed as

$$\mathbf{z}_t = \mathbf{z}_{t-1} + f(\mathbf{z}_{t-1})\Delta t + \mathbf{w}_t,$$

where f is a deterministic function and $\mathbf{w}_t$ is random noise.

Shuffling the terms, we get

$$\frac{\mathbf{z}_t - \mathbf{z}_{t-1}}{\Delta t} = f(\mathbf{z}_{t-1}) + \frac{\mathbf{w}_t}{\Delta t},$$

which, in the limit as $\Delta t \rightarrow 0$, gives us a continuous-time model

$$\frac{d\mathbf{z}(t)}{dt} = f(\mathbf{z}(t)) + \frac{d\mathbf{w}(t)}{dt},$$

where $\mathbf{z}(t)$ is the state at time t and $\mathbf{w}(t)$ is continuous-time noise.

Without the noise, we get a **deterministic** system, an **ordinary differential equation** (ODE),

$$\frac{dz(t)}{dt} = f(z(t)).$$

Example. Exponential decay to an equilibrium point μ ,

$$\frac{dz(t)}{dt} = -\kappa(z(t) - \mu), \quad \text{with solution} \quad z(t) = \mu + (z(0) - \mu)e^{-\kappa t}.$$

The state moves towards μ from either side, and is fully determined by $z(0)$.

Brownian motion

To add stochasticity to the ODE, we need a continuous-time stochastic process that can model random noise.

We can model the noise term $\mathbf{w}(t)$ as a standard Brownian motion (Wiener process) $W(t)$, which has the following properties:

- $W(0) = 0$.
- For $0 \leq s < t$, the increment $W(t) - W(s) \sim \mathcal{N}(0, t - s)$.
- Increments over disjoint intervals are independent.
- $W(t)$ is continuous in t .

A vector $\mathbf{W}(t)$ has independent Brownian components.

Adding Brownian motion to the ODE, scaled by a **diffusion term** g , gives a **stochastic differential equation** (SDE),

$$d\mathbf{z}(t) = f(\mathbf{z}(t)) dt + g(\mathbf{z}(t)) d\mathbf{W}(t).$$

Over an infinitesimal interval dt , the drift $f dt$ is the deterministic change of the state, and the diffusion $g d\mathbf{W}$ its random change.

The differential form is needed because Brownian motion is nowhere differentiable: $d\mathbf{W}(t)/dt$, white noise, is only symbolic.

Example: Ornstein-Uhlenbeck process

Recall our animal movement example in discrete time:

$$\mathbf{z}_t = \mathbf{z}_{t-1} - \kappa(\mathbf{z}_{t-1} - \boldsymbol{\mu})\Delta t + \mathbf{w}_t,$$

where $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \Delta t \mathbf{I})$.

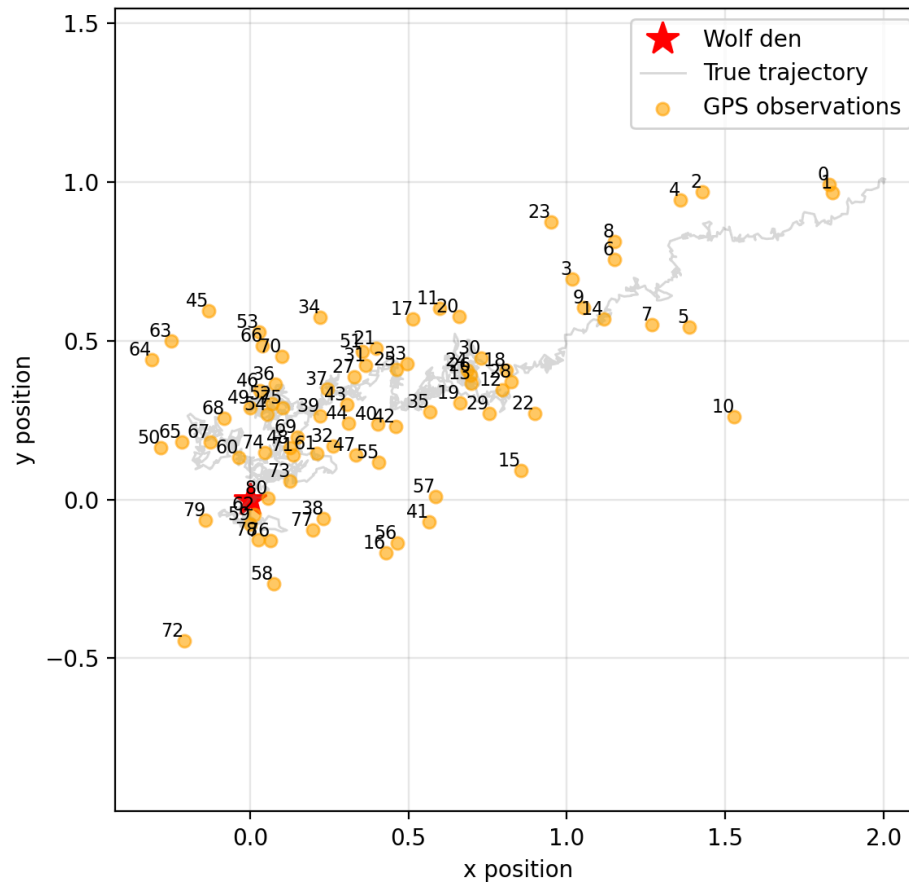
In continuous time, this becomes the SDE

$$d\mathbf{z}(t) = -\kappa(\mathbf{z}(t) - \boldsymbol{\mu})dt + \sigma d\mathbf{W}(t),$$

where $\kappa > 0$ is the strength of attraction to the home location $\boldsymbol{\mu}$ and σ is the diffusion coefficient.

This process is known as the **Ornstein-Uhlenbeck process**, which describes a mean-reverting behavior with Gaussian noise.

The discrete-time model is the Euler-Maruyama discretization of the OU process with step size Δt .



The true trajectory behind the GPS observations: an Ornstein-Uhlenbeck process, simulated on a fine grid and observed every 0.25 time units.

Exact discretization

Observed at times $t_1 < t_2 < \dots$, possibly irregular, the OU process can be discretized exactly. With $\Delta_i = t_i - t_{i-1}$,

$$p(\mathbf{z}(t_i) \mid \mathbf{z}(t_{i-1})) = \mathcal{N} \left(\boldsymbol{\mu} + e^{-\kappa \Delta_i} (\mathbf{z}(t_{i-1}) - \boldsymbol{\mu}), \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa \Delta_i}) \mathbf{I} \right).$$

This is a linear Gaussian transition, with $\mathbf{A}_i = e^{-\kappa \Delta_i} \mathbf{I}$ and $\mathbf{b}_i = (1 - e^{-\kappa \Delta_i}) \boldsymbol{\mu}$. The Kalman filter and smoother apply unchanged, with matrices that change with the time step. The figures of this lecture use it, at $\Delta = 0.25$.

When to use continuous vs discrete time?

Continuous time suits observations at irregular intervals, mechanistic models, and parameters that are rates or time constants.

Discrete time suits regular sampling and models without a mechanistic interpretation, and is simpler to compute with.

A common approach is to model in continuous time, for interpretability, and to discretize for computation.

